Tutorial on the Robust Interior Point Method

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Abstract

We give a short, self-contained proof of the interior point method and its robust version.

1 Introduction

Consider the primal linear program

$$\min_{\mathbf{A}x=b,x\in\mathbb{R}^n_{\geq 0}} c^\top x \tag{P}$$

and its dual

$$\max_{\mathbf{A}^\top y+s=c,s\in\mathbb{R}^n_{\geq 0}} b^\top y \tag{D}$$

where $\mathbf{A}\in\mathbb{R}^{d\times n}$, $\mathbb{R}_{\geq 0}$ denotes the set of nonnegative real numbers, and vector inequalities are coordinatewise. The feasible regions for the two programs are

$$\mathcal{P}=\{x\in\mathbb{R}^n_{\geq 0}:\mathbf{A}x=b\} \text{ and } \mathcal{D}=\{s\in\mathbb{R}^n_{\geq 0}:\mathbf{A}^\top y+s=c \text{ for some } y\}.$$

We define their interiors:

$$\mathcal{P}^\circ=\{x\in\mathbb{R}^n_{>0}:\mathbf{A}x=b\} \text{ and } \mathcal{D}^\circ=\{s\in\mathbb{R}^n_{>0}:\mathbf{A}^\top y+s=c \text{ for some } y\}.$$

One key property about the primal and dual linear program is that dual objective $b^\top y$ is a lower bound of the primal object $c^\top y$ and that $x^\top s$ measure the gap between two objective.

To motivate the main idea of the interior point method, we recall the optimality condition for linear programs.

Lemma 1 (Duality Gap)**.** *For any $x\in\mathcal{P}$ and any dual feasible pair (y,s) with $\mathbf{A}^\top y+s=c$, the duality gap satisfies $c^\top x-b^\top y=x^\top s$. In particular, we have*

$$\max_{\mathbf{A}^\top\overline{y}+\overline{s}=c,\overline{s}\in\mathbb{R}^n_{\geq 0}} b^\top\overline{y}-x^\top s\leq b^\top y\leq c^\top x\leq \min_{\mathbf{A}x=b,x\in\mathbb{R}^n_{\geq 0}} c^\top\overline{x}+x^\top s.$$

Proof. Using $\mathbf{A}x=b$ and $\mathbf{A}^\top y+s=c$, we can compute the duality gap as follows

$$c^\top x-b^\top y=c^\top x-(\mathbf{A}x)^\top y=c^\top x-x^\top(\mathbf{A}^\top y)=x^\top s.$$

Since $x,s\geq 0$, this shows that

$$c^\top x\geq b^\top y.$$

Optimize over all x and y, we have the weak duality

$$\min_{\mathbf{A}x=b,x\in\mathbb{R}^n_{\geq 0}} c^\top\overline{x}\geq \max_{\mathbf{A}^\top y+s=c,s\in\mathbb{R}^n_{\geq 0}} b^\top y.$$

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Using this, we have

$$c^\top x = b^\top y + x^\top s \leq \max_{\mathbf{A}^\top y+s=c, s\in\mathbb{R}^n_{\geq 0}} b^\top y + x^\top s \leq \min_{\mathbf{A}x=b, x\in\mathbb{R}^n_{\geq 0}} c^\top \overline{x} + x^\top s.$$

The another direction is similar. $\square$

The main implication of Lemma 1 is that any feasible (x,s) with small $x^\top s$ is a nearly optimal solution of the linear program. This leads us to primal-dual algorithms in which we start with a feasible primal-dual solution pair (x,s) and iteratively update the solution to decrease the duality gap $x^\top s$.

2 Interior Point Method

In this section, we discuss the classical short-step interior point method. For two vectors a,b, we use ab to denote the vector with components $(ab)_i = a_i b_i$ and a/b to denote the vector with components a_i/b_i. For a scalar $t\in\mathbb{R}$, we let $t\mathbf{1}$ denote the vector with all coordinates equal to t.

Definition 2 (Central Path)**.** We define the central path $(x_t, s_t)\in\mathcal{P}^\circ\times\mathcal{D}^\circ$ by $x_t s_t = t\mathbf{1}$. We say x_t is on the central path of (P) at t.

The algorithm maintains a pair $(x,s)\in\mathcal{P}^\circ\times\mathcal{D}^\circ$ and a scalar $t>0$ satisfying the invariant

$$\|\frac{xs}{t} - \mathbf{1}\|_2 \leq \frac{1}{4}.$$

Note that the deviation from the central path is measured in ℓ_2 norm. In each step, the algorithm decreases t by a factor of $1-\Omega(n^{-1/2})$ while maintaining the invariant.

2.1 Basic Property of a Step

To see why there is a pair $(x,s)\in\mathcal{P}^\circ\times\mathcal{D}^\circ$ satisfying the invariant, we prove the following generalization.

Lemma 3 (Quadrant Representation of Primal-Dual)**.** *Suppose $\mathcal{P}^\circ$ is non-empty and $\mathcal{P}$ is bounded. For any positive vector $\mu\in\mathbb{R}^n_{>0}$, there is a unique pair $(x_\mu, s_\mu)\in\mathcal{P}^\circ\times\mathcal{D}^\circ$ such that $x_\mu s_\mu = \mu$. Furthermore, $x_\mu = \arg\min_{x\in\mathcal{P}} f_\mu(x)$ where*

$$f_\mu(x) = c^\top x - \sum_{i=1}^n \mu_i \ln x_i.$$

Proof. Fix $\mu\in\mathbb{R}^n_{>0}$. We define $x_\mu = \arg\min_{x\in\mathcal{P}} f_\mu(x)$ and prove that $(x_\mu,s_\mu)\in\mathcal{P}^\circ\times\mathcal{D}^\circ$ with $x_\mu s_\mu = \mu$ for some s_μ. Since $\mathcal{P}^\circ$ is non-empty, $\mathcal{P}$ is bounded, and f_μ is strictly convex, such an x_μ exists. Furthermore, since $f_\mu(x)\to+\infty$ as $x_i\to 0$ for any i, we have that $x_\mu\in\mathcal{P}^\circ$.

By the KKT optimality condition for f_μ, there is a vector y such that

$$\nabla f_\mu(x_\mu) = c - \frac{\mu}{x_\mu} = \mathbf{A}^\top y.$$

Define $s_\mu = \frac{\mu}{x_\mu}$. Then one can check that $s_\mu\in\mathcal{D}^\circ$ and $x_\mu s_\mu = \mu$.

For the uniqueness, if $(x,s)\in\mathcal{P}^\circ\times\mathcal{D}^\circ$ and $xs=\mu$, then x satisfies the optimality condition for f_μ. Since f_μ is strictly convex, such x must be unique. $\square$

Lemma 3 shows that any point in $\mathcal{P}^\circ\times\mathcal{D}^\circ$ is uniquely represented by a positive vector μ. Interior point methods move μ uniformly to 0 while maintaining the corresponding x_μ. Now we discuss how to find (x_μ,s_μ) from a nearby interior feasible point (x,s); namely, how to move (x,s) to $(x+\delta_x, s+\delta_s)$ so that the result satisfies the system

$$\begin{aligned}
(x+\delta_x)(s+\delta_s) &= \mu,\\
\mathbf{A}(x+\delta_x) &= b,\\
\mathbf{A}^\top(y+\delta_y) + (s+\delta_s) &= c,\\
(x+\delta_x, s+\delta_s) &\in \mathbb{R}^{2n}_{>0}.
\end{aligned}$$

Although the system above involves y, our approximate solution does not need to know y. By ignoring the inequality constraint and the second-order term $\delta_x \delta_s$ in the first equation above, and using $\mathbf{A}x = b$ and $\mathbf{A}^\top y + s = c$ we can simplify the system:

$$\begin{aligned} xs + \mathbf{S}\delta_x + \mathbf{X}\delta_s &= \mu, \\ \mathbf{A}\delta_x &= 0, \\ \mathbf{A}^\top \delta_y + \delta_s &= 0, \end{aligned} \tag{2.1}$$

where $\mathbf{X}$ and $\mathbf{S}$ are the diagonal matrices with diagonals x and s, respectively. In the following lemma, we show how to write the step above using a projection matrix ($\mathbf{P}^2 = \mathbf{P}$).

Lemma 4. *Suppose that* $\mathbf{A}$ *has full row rank and* $(x, s) \in \mathcal{P}^\circ \times \mathcal{D}^\circ$. *Then, the unique solution for the linear system (2.1) is given by*

$$\begin{aligned} \mathbf{X}^{-1}\delta_x &= (\mathbf{I} - \mathbf{P})(\delta_\mu/(xs)), \\ \mathbf{S}^{-1}\delta_s &= \mathbf{P}(\delta_\mu/(xs)) \end{aligned}$$

where $\delta_\mu = \mu - xs$, *the division by* xs *is coordinatewise, and* $\mathbf{P} = \mathbf{S}^{-1}\mathbf{A}^\top(\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top)^{-1}\mathbf{A}\mathbf{X}$.

Proof. Note that the step satisfies $\mathbf{S}\delta_x + \mathbf{X}\delta_s = \delta_\mu$. Multiplying both sides by $\mathbf{A}\mathbf{S}^{-1}$ and using $\mathbf{A}\delta_x = 0$, we have

$$\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\delta_s = \mathbf{A}\mathbf{S}^{-1}\delta_\mu.$$

Now we use that $\mathbf{A}^\top \delta_y + \delta_s = 0$ and get

$$\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top \delta_y = -\mathbf{A}\mathbf{S}^{-1}\delta_\mu.$$

Since $\mathbf{A} \in \mathbb{R}^{d \times n}$ has full row rank and $\mathbf{S}^{-1}\mathbf{X}$ is invertible, we have that $\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top$ is invertible. Hence,

$$\delta_y = -(\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top)^{-1}\mathbf{A}\mathbf{S}^{-1}\delta_\mu \qquad \text{and} \qquad \delta_s = \mathbf{A}^\top(\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top)^{-1}\mathbf{A}\mathbf{S}^{-1}\delta_\mu.$$

Putting this into $\mathbf{S}\delta_x + \mathbf{X}\delta_s = \delta_\mu$, we have

$$\delta_x = \mathbf{S}^{-1}\delta_\mu - \mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top(\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top)^{-1}\mathbf{A}\mathbf{S}^{-1}\delta_\mu.$$

The result follows from the definition of $\mathbf{P}$. □

2.2 Lower Bounding Step Size

The efficiency of interior point methods depends on how large a step we can take while staying within the domain. In the rest of this subsection, we write $\mu = xs$ for the current complementarity vector. We first study the step operators $(\mathbf{I} - \mathbf{P})$ and $\mathbf{P}$. The following lemma implies that $\mathbf{P}$ is a nearly orthogonal projection matrix when μ is close to a multiple of the all-ones vector. Hence, the relative changes of $\mathbf{X}^{-1}\delta_x$ and $\mathbf{S}^{-1}\delta_s$ are essentially the orthogonal decomposition of the relative step δ_μ/μ. For two vectors u, v, we define $\|u\|_v \stackrel{\text{def}}{=} \sqrt{u^\top \mathrm{Diag}(v) u}$ to be the norm defined by v.

Lemma 5. *Under the assumption in Lemma 4,* $\mathbf{P}$ *is a projection matrix such that* $\|\mathbf{P}v\|_\mu \leq \|v\|_\mu$ *for any* $v \in \mathbb{R}^n$. *Similarly, we have that* $\|(\mathbf{I} - \mathbf{P})v\|_\mu \leq \|v\|_\mu$.

Proof. $\mathbf{P}$ is a projection because $\mathbf{P}^2 = \mathbf{P}$. Define the orthogonal projection matrix

$$\mathbf{P}_{\text{orth}} = \mathbf{S}^{-1/2}\mathbf{X}^{1/2}\mathbf{A}^\top(\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top)^{-1}\mathbf{A}\mathbf{X}^{1/2}\mathbf{S}^{-1/2},$$

then we have

$$\begin{aligned} \|\mathbf{P}v\|_\mu^2 &= v^\top \mathbf{X}\mathbf{A}^\top(\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top)^{-1}\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{S}\mathbf{S}^{-1}\mathbf{A}^\top(\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top)^{-1}\mathbf{A}\mathbf{X}v \\ &= v^\top \mathbf{S}^{1/2}\mathbf{X}^{1/2}\mathbf{P}_{\text{orth}}\mathbf{S}^{1/2}\mathbf{X}^{1/2}v \\ &\leq v^\top \mathbf{S}^{1/2}\mathbf{X}^{1/2}\mathbf{S}^{1/2}\mathbf{X}^{1/2}v = \|v\|_\mu^2. \end{aligned}$$

The calculation for $\|(\mathbf{I} - \mathbf{P})v\|_\mu$ is similar. □

Next we give a lower bound on the largest feasible step size.

Lemma 6. *We have that* $\|\mathbf{X}^{-1}\delta_x\|_\infty^2 \le \frac{1}{\min_i \mu_i}\|\delta_\mu/\mu\|_\mu^2$ *and* $\|\mathbf{S}^{-1}\delta_s\|_\infty^2 \le \frac{1}{\min_i \mu_i}\|\delta_\mu/\mu\|_\mu^2$. *In particular, if* $\|\delta_\mu/\mu\|_\mu^2 < \min_i \mu_i$, *we have* $(x+\delta_x, s+\delta_s) \in \mathcal{P}^\circ \times \mathcal{D}^\circ$

Proof. For $\|\mathbf{X}^{-1}\delta_x\|_\infty$, we have $\min_i \mu_i \|\mathbf{X}^{-1}\delta_x\|_\infty^2 \le \|\mathbf{X}^{-1}\delta_x\|_\mu^2$ and hence

$$\|\mathbf{X}^{-1}\delta_x\|_\infty^2 \le \frac{1}{\min_i \mu_i}\|\mathbf{X}^{-1}\delta_x\|_\mu^2 = \frac{1}{\min_i \mu_i}\|(\mathbf{I}-\mathbf{P})(\delta_\mu/\mu)\|_\mu^2 \le \frac{1}{\min_i \mu_i}\|\delta_\mu/\mu\|_\mu^2.$$

The proof for $\|\mathbf{S}^{-1}\delta_s\|_\infty$ is similar.

Hence, if $\|\delta_\mu/\mu\|_\mu^2 < \min_i \mu_i$, we have that $\|\mathbf{X}^{-1}\delta_x\|_\infty < 1$ and $\|\mathbf{S}^{-1}\delta_s\|_\infty < 1$, i.e., $|\delta_{x,i}| < |x_i|$ and $|\delta_{s,i}| < |s_i|$ for all i. Therefore, $x+\delta_x > 0$ and $s+\delta_s > 0$ are feasible. ☐

To decrease μ uniformly, we set $\delta_\mu = -h\mu$ for some step size h. To ensure the feasibility, we need $\|\delta_\mu/\mu\|_\mu^2 \le \min_i \mu_i$ and this gives the maximum step size

$$h = \sqrt{\frac{\min_i \mu_i}{\sum_i \mu_i}}. \tag{2.2}$$

Note that the above quantity is maximized at $h = n^{-1/2}$ when μ has all equal coordinates.

2.3 Staying within a small ℓ_2 distance

The step size (2.2) is maximized when μ is a constant vector. A natural approach is therefore to keep μ as a vector close in ℓ_2 norm to a multiple of the all-ones vector. This motivates the following algorithm:

Algorithm 1: `L2Step`$(\mathbf{A}, x, s, t_{\text{start}}, t_{\text{end}})$

Define $\mathbf{P}_{x,s} = \mathbf{S}^{-1}\mathbf{A}^\top(\mathbf{A}\mathbf{S}^{-1}\mathbf{X}\mathbf{A}^\top)^{-1}\mathbf{A}\mathbf{X}$.
Invariant: $(x,s) \in \mathcal{P}^\circ \times \mathcal{D}^\circ$ and $\|xs - t\|_2 \le \frac{t}{4}$.
Let $t = t_{\text{start}}$ and $h = 1/(16\sqrt{n})$, where n is the number of columns in $\mathbf{A}$.
repeat
  Let $t' = \max(t/(1+h), t_{\text{end}})$.
  Let $\mu = xs$ and $\delta_\mu = t' - \mu$.
  Let $\delta_x = \mathbf{X}(\mathbf{I} - \mathbf{P}_{x,s})(\delta_\mu/\mu)$ and $\delta_s = \mathbf{S}\mathbf{P}_{x,s}(\delta_\mu/\mu)$.
  Set $x \leftarrow x + \delta_x$, $s \leftarrow s + \delta_s$ and $t \leftarrow t'$.
until $t = t_{\text{end}}$;
Return (x,s).

Note that the algorithm requires some initial point (x,s) close to the central path and we will show how to get this in the next section (by changing the linear program temporarily).

First, we show that the invariant is maintained in each step. The conclusion that the distance is less than $t/6$ is needed in Section A, where we call `L2Step` on a modified LP and then prove that the result is close to the central path for the original LP.

Lemma 7. *Suppose that the input satisfies* $(x,s) \in \mathcal{P}^\circ \times \mathcal{D}^\circ$ *and* $\|xs - t_{\text{start}}\|_2 \le \frac{t_{\text{start}}}{4}$. *Then, the algorithm* `L2Step` *maintains* $(x,s) \in \mathcal{P}^\circ \times \mathcal{D}^\circ$ *and* $t > 0$ *such that* $\|xs - t\|_2 \le \frac{t}{6}$.

Proof. We prove by induction that $\|xs - t\|_2 \le \frac{t}{6}$ after each step. Note that the input satisfies $\|xs - t\|_2 \le \frac{t}{4}$.

Let $x' = x + \delta_x$, $s' = s + \delta_s$ and let t' be as defined in the algorithm. Note that

$$x's' - t' = (x+\delta_x)(s+\delta_s) - t' = \mu + \mathbf{S}\delta_x + \mathbf{X}\delta_s + \delta_x\delta_s - t'.$$

Lemma 4 shows that $\mathbf{S}\delta_x + \mathbf{X}\delta_s = t' - \mu$. Hence, we have

$$x's' - t' = \delta_x\delta_s = \mathbf{X}^{-1}\delta_x \cdot \mathbf{S}^{-1}\delta_s \cdot \mu.$$

Using this, we have

$$\|x's' - t'\|_2 \le \|\mu^{1/2}\mathbf{X}^{-1}\delta_x\|_2 \|\mu^{1/2}\mathbf{S}^{-1}\delta_s\|_2 = \|\mathbf{X}^{-1}\delta_x\|_\mu \|\mathbf{S}^{-1}\delta_s\|_\mu \le \|\delta_\mu/\mu\|_\mu^2.$$

Here we used Lemma 5 at the end.

Using $t' - \mu = \frac{t'}{t}(t-\mu) + (\frac{t'}{t} - 1)\mu$, we have

$$\|\delta_\mu/\mu\|_\mu = \|\frac{t'}{t}\frac{t-\mu}{\mu} + (\frac{t'}{t} - 1)\|_\mu \le \frac{t'}{t}\|xs - t\|_{\mu^{-1}} + \|\frac{t'}{t} - 1\|_\mu.$$

Since $\|\mu - t\|_2 \le \frac{t}{4}$, we have $\min_i \mu_i \ge \frac{3t}{4}$ and $\max_i \mu_i \le \frac{5}{4}t$. Using $|\frac{t'}{t} - 1| \le h = \frac{1}{16\sqrt{n}}$, we have

$$\|\delta_\mu/\mu\|_\mu \le \frac{t'}{t}\sqrt{\frac{4}{3t}}\|xs - t\|_2 + h\sqrt{\frac{5}{4}tn} \le \sqrt{\frac{t}{12}} + h\sqrt{\frac{5}{4}tn} \le 0.36\sqrt{t}.$$

Hence, we have $\|x's' - t'\|_2 \le \|\delta_\mu/\mu\|_\mu^2 \le 0.13t \le t'/6$. Furthermore, $\|\delta_\mu/\mu\|_\mu^2 < \min_i \mu_i$, which implies (x', s') is feasible (Lemma 6). $\square$

Note that the lemma above only concludes that the output is close to the central path. To upper bound the error, we can apply Lemma 1 which shows the duality gap is equal to $x^\top s$.

2.4 Solving LP Approximately and Exactly

Here we discuss how to get a feasible interior point close to the central path by modifying the linear program. The runtime of an interior point method depends on how degenerate the linear program is.

Definition 8. We define the following parameters for the linear program $\min_{\mathbf{A}x=b, x\ge 0} c^\top x$:

1. Inner radius r: There exists an $x \in \mathcal{P}$ such that $x_i \ge r$ for all $i \in [n]$.
2. Outer radius R: For any $x \ge 0$ with $\mathbf{A}x = b$, we have that $\|x\|_2 \le R$.
3. Lipschitz constant L: $\|c\|_2 \le L$.

Since `L2Step` requires a feasible point near the central path, we modify the linear program to make it happen. To satisfy the constraint $\mathbf{A}x = b$, we start the algorithm by taking a least-squares solution of the constraint $\mathbf{A}x = b$. Since it can be negative, we write the variable $x = x^+ - x^-$ with both $x^+, x^- \ge 0$. We put a large cost vector on x^- to ensure the solution is roughly the same. The crux of the proof is that if we optimize this new program well enough, we will have $x^+ - x^- > 0$ and hence $x^+ - x^-$ gives a good starting point of the original program. For technical reasons, we need to put an extra constraint $1^\top x^+ \le \Lambda$ for some Λ to ensure the problem is bounded.

Theorem 9. *Given a linear program $\min_{\mathbf{A}x=b, x\in\mathbb{R}^n_{\ge 0}} c^\top x$ with inner radius r, outer radius R and Lipschitz constant L and any $0 \le \eta \le \frac{1}{3}$. Consider the modified linear program by*

$$\min_{(x^+,x^-,x^\theta)\in\mathcal{P}_\eta} \langle c^+, x^+\rangle + \langle c^-, x^-\rangle$$

with

$$\mathcal{P}_\eta = \left\{(x^+, x^-, x^\theta) \in \mathbb{R}^{2n+1}_{\ge 0} : \ \mathbf{A}(x^+ - x^-) = b, \sum_{i=1}^n x_i^+ + x^\theta = \widetilde{b}\right\}$$

where $\overline{R} = 6R/\eta$, $\overline{t} = 4000n^2R/(\eta^2 r)\cdot LR$, $\overline{x}^+ = \overline{t}/(c + \overline{t}/\overline{R})$, $\overline{x}^- = \overline{x}^+ - \mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}b$, $c^+ = c$, $c^- = t/\overline{x}^-$ and $\widetilde{b} = \sum_{i=1}^n \overline{x}_i^+ + \overline{R}$. Then, we have that

1. *$(\overline{x}^+, \overline{x}^-, \overline{R})$ is on the central path of the modified program at $\overline{t}$.*
2. *For any primal $x \stackrel{\text{def}}{=} (x^+, x^-, x^\theta) \in \mathcal{P}_\eta$ and dual $s \stackrel{\text{def}}{=} (s^+, s^-, s^\theta)$ such that $\frac{5}{6}LR \le x_i s_i \le \frac{7}{6}LR$, we have that*

   $$(x^+ - x^-, s^+ - s^\theta) \in \mathcal{P} \times \mathcal{D}$$

   and that $x_i^- \le \eta x_i^+$ and $s^\theta \le \eta s_i^+$ for all i.

Proof. Since the proof is not illuminating, we defer it to Appendix A. $\square$

Now we state our main algorithm.

Algorithm 2: `SlowSolveLP`$(\mathbf{A}, b, c, \delta)$

Assumption: the linear program has inner radius r, outer radius R and Lipschitz constant L.
Let $\epsilon = 1/(48\sqrt{n})$, $\overline{R} = 6R/\eta$, $\overline{t} = 4000n^2R/(\eta^2 r) \cdot LR$.
`// Define the modified program` $\min_{\overline{\mathbf{A}}x=\overline{b}} \overline{c}^\top x$ `by Theorem 9 with parameters` $\overline{R}$ `and` $\overline{t}$.
Let $\overline{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & -\mathbf{A} & 0 \\ 1_n^\top & 0 & 1 \end{bmatrix}$, $\overline{c} = (c, c^-, 0)$, $\overline{b} = (b, \widetilde{b})$ where c^- and $\widetilde{b}$ are defined in Definition 9.
Let $(\overline{x}, \overline{s})$ be the explicit central path at $\overline{t}$ given by Theorem 9.
$(\overline{x}, \overline{s}) \leftarrow$ `L2Step`$(\overline{\mathbf{A}}, \overline{x}, \overline{s}, \overline{t}, LR)$.
$(x, s) \leftarrow (\overline{x}^+ - \overline{x}^-, \overline{s}^+ - \overline{s}^\theta)$ where $\overline{x} = (\overline{x}^+, \overline{x}^-, \overline{x}^\theta)$ and $\overline{s} = (\overline{s}^+, \overline{s}^-, \overline{s}^\theta)$.
$(x_{\text{end}}, s_{\text{end}}) =$ `L2Step`$(\mathbf{A}, x^+ - x^-, s^+ - s^\theta, LR, t_{\text{end}})$ with $t_{\text{end}} = \delta LR/(2n)$.
Return x_{end}.

Theorem 10. *Consider a linear program $\min_{\mathbf{A}x=b,x\geq 0} c^\top x$ with n variables and d constraints. Assume the linear program has inner radius r, outer radius R and Lipschitz constant L(see Definition 8). Then,* `SlowSolveLP` *outputs x such that*

$$\begin{aligned} c^\top x &\leq \min_{\mathbf{A}x=b,x\geq 0} c^\top x + \delta LR, \\ \mathbf{A}x &= b, \\ x &\geq 0. \end{aligned}$$

The algorithm takes $O(\sqrt{n}\log(nR/(\delta r)))$ Newton steps (defined in (2.1)).

If we further assume that the solution $x^* = \arg\min_{\mathbf{A}x=b,x\geq 0} c^\top x$ is unique and that $c^\top x \geq c^\top x^* + \zeta LR$ for any other vertex x of $\{\mathbf{A}x = b, x \geq 0\}$ for some $\zeta > \delta \geq 0$, then we have that $\|x - x^*\|_2 \leq \frac{2\delta R}{\zeta}$.

Proof. By Theorem 9, the point $(\overline{x}, \overline{s})$ is on the central path of the modified program at $\overline{t}$. After the first call of `L2Step`, Lemma 7 shows that `L2Step` returns $(\overline{x}, \overline{s})$ such that $\|\overline{x}\overline{s} - t\|_2 \leq \frac{t}{6}$ with $t = LR$.

Theorem 9 shows that $(x, s) = (\overline{x}^+ - \overline{x}^-, \overline{s}^+ - \overline{s}^\theta) \in \mathcal{P} \times \mathcal{D}$ and that $(1-\eta)\overline{x}_i^+ \leq x_i \leq \overline{x}_i^+$ and $(1-\eta)\overline{s}_i^+ \leq s_i \leq \overline{s}_i^+$. Since $\eta = \frac{1}{48\sqrt{n}}$ and $\|\overline{x}^+\overline{s}^+ - t\|_2 \leq \frac{t}{6}$ with $t = LR$, we have that $\|xs - t\|_2 \leq \frac{t}{4}$. This verifies the condition for the second call of `L2Step`.

After the second call of `L2Step`, Lemma 7 shows that `L2Step` returns $(x_{\text{end}}, s_{\text{end}})$ such that $\|x_{\text{end}}s_{\text{end}} - t_{\text{end}}\|_2 \leq \frac{t_{\text{end}}}{6}$ with $t_{\text{end}} = \delta LR/(2n)$. Hence, Lemma 1 shows that

$$c^\top x_{\text{end}} \leq \min_{\mathbf{A}x=b,x\geq 0} c^\top x + x_{\text{end}}^\top s_{\text{end}} \leq \min_{\mathbf{A}x=b,x\geq 0} c^\top x + 2t_{\text{end}}n \leq \min_{\mathbf{A}x=b,x\geq 0} c^\top x + \delta LR.$$

For the runtime, note that `L2Step` decreases t by a factor of $1 - \Omega(n^{-1/2})$ in each step. Hence, the first call takes $O(\sqrt{n}\log(nR/r))$ Newton steps and the second call takes $O(\sqrt{n}\log(n/\delta))$ Newton steps.

For the last conclusion, we assume $\delta \leq \zeta$ and let $\mathcal{P}(\zeta) = \mathcal{P} \cap \{c^\top x \leq c^\top x^* + tLR\}$. Note that $\mathcal{P}(\zeta)$ is a cone at x^* (because there is no vertex other than x^* with value less than $c^\top x^* + \zeta LR$). Hence, we have $\mathcal{P}(\delta) - x^* = \frac{\delta}{\eta}(\mathcal{P}(\zeta) - x^*)$. Since $x \in \mathcal{P}_\delta$, we have that

$$\|x - x^*\|_2 \leq \frac{\delta}{\zeta}\text{diameter}(\mathcal{P}(\zeta) - x^*) \leq \frac{2\delta R}{\zeta}.$$

□

If we know the solution of the linear program is integral or rational with some bound on the number of bits, then getting a solution close enough to x^* allows us to round the solution to an integral solution. Therefore, the last conclusion of the theorem above gives us an exact linear program algorithm assuming $\mathbf{A}, b, c$ are integral and bounded. The uniqueness assumption can be achieved by perturbing the cost vector by a random vector (e.g., using the "isolation" lemma [10, Lemma 4]).

Exercise 11. Make the perturbation deterministic while preserving solutions.

3 Robust Interior Point Method

To improve the interior point method, one can either improve the number of steps $\widetilde{O}(\sqrt{n})$ or the cost per step. The first is a major open problem. In this note, we focus on the latter question. Recall from (2.1) that the linear system we solve in each step is of the form

$$\begin{aligned}\mathbf{S}\delta_x + \mathbf{X}\delta_s &= \delta_\mu,\\ \mathbf{A}\delta_x &= 0,\\ \mathbf{A}^\top\delta_y + \delta_s &= 0.\end{aligned} \tag{3.1}$$

In each step, x, s and δ_μ in the equation above change relatively by vectors with bounded ℓ_2 norm. Thus, only a few coordinates change substantially in each step. To take advantage of this, the robust interior point method contains two new components: (1) an analysis of convergence when we only solve the linear system approximately (Section 3.1), and (2) a method for maintaining the solution throughout the iteration (Section 3.2 and Section 3.4).

3.1 Staying within small ℓ_∞ distance

In the preceding description and analysis, we assumed that we computed each step of the interior point method precisely. However, it may suffice to compute steps approximately since our goal is only to stay close to the central path. This could have significant computational advantages.

To make the interior point method robust to noise in the updates to x and s, we need the method to work under a larger neighborhood than that given by the Euclidean norm ($\|xs-t\|_2 \leq \frac{t}{4}$). We cannot increase the radius of the ℓ_2 ball because we need the neighborhood to lie strictly inside the feasible region. One natural alternative choice of distance and potential would be a higher norm, $\|xs-t\|_q^q$. However, analyzing the step δ_μ that minimizes $\|\mu+\delta_\mu-t\|_q^q$ involves many cases. Instead, we use the potential

$$\Phi(r) = \sum_{i=1}^{n}\cosh(\lambda r_i) = \sum_{i=1}^{n}\frac{(e^{\lambda r_i}+e^{-\lambda r_i})}{2}. \tag{3.2}$$

with $r = \frac{xs-t}{t}$ for some scalar $\lambda = \Theta(\log n)$. This potential induces the following algorithm, whose steps use $\delta_\mu \approx -c\nabla\Phi(\frac{xs-t}{t})$.

Algorithm 3: `RobustStep`$(\mathbf{A}, x, s, t_{\text{start}}, t_{\text{end}})$

Define $\Phi(r)$ according to (3.2) with $\lambda = 16\log(40n)$.
Invariant: $(x,s)\in\mathcal{P}^\circ\times\mathcal{D}^\circ$ and $\Phi((xs-t)/t)\leq 16n$.
Let $t = t_{\text{start}}$, $h = 1/(128\lambda\sqrt{n})$, and let n be the number of columns in $\mathbf{A}$.
repeat
  Pick $\overline{x}$, $\overline{s}$ and $\overline{r}$ such that $\|\ln\overline{x}-\ln x\|_\infty \leq \frac{1}{48}$, $\|\ln\overline{s}-\ln s\|_\infty\leq\frac{1}{48}$ and $\|\overline{r}-\frac{xs-t}{t}\|_\infty\leq\frac{1}{48\lambda}$.
  Let $t' = \max(t/(1+h), t_{\text{end}})$, $\overline{\delta}_\mu = -\frac{t'}{32\lambda}\frac{\overline{g}}{\|\overline{g}\|_2}$, $\overline{g} = \nabla\Phi(\overline{r})$.
  Find δ_x, δ_s such that

$$\begin{aligned}\overline{\mathbf{S}}\delta_x + \overline{\mathbf{X}}\delta_s &= \overline{\delta}_\mu,\\ \mathbf{A}\delta_x &= 0,\\ \mathbf{A}^\top\delta_y + \delta_s &= 0.\end{aligned} \tag{3.3}$$

  Set $x \leftarrow x+\delta_x$, $s\leftarrow s+\delta_s$ and $t\leftarrow t'$.
until $t = t_{\text{end}}$;
Return (x,s).

We begin with useful facts about Φ.

Lemma 12. *Define $\Phi(r)$ according to (3.2). For any $r\in\mathbb{R}^n$, we have that $\|r\|_\infty \leq \frac{\log(2\Phi(r))}{\lambda}$ and $\|\nabla\Phi(r)\|_2 \geq \frac{\lambda}{\sqrt{n}}(\Phi(r)-n)$. Moreover, if $\Phi(r)\geq 4n$ and $\|\delta\|_\infty\leq\frac{1}{5\lambda}$, we have*

$$\|\nabla\Phi(r+\delta)-\nabla\Phi(r)\|_2 \leq \frac{1}{3}\|\nabla\Phi(r)\|_2.$$

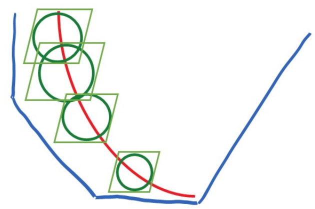

Figure 3.1: The Central Path. The standard method keeps the current point within a relative ℓ_2-norm ball, while it suffices to use a larger ℓ_∞-norm ball. The robust method provides a bridge to the latter via the cosh ball defined by Φ.

Proof. For every i, we have $\Phi(r) \geq \frac{1}{2}e^{\lambda|r_i|}$, and hence $\|r\|_\infty \leq \frac{\log(2\Phi(r))}{\lambda}$.

For the second claim, using that $\nabla\Phi(r) = (\lambda\sinh(\lambda r_i))_{i=1}^n$, we have

$$\begin{aligned}
\|\nabla\Phi(r)\|_2 &= \lambda\sqrt{\sum_{i=1}^n \sinh^2(\lambda r_i)} = \lambda\sqrt{\sum_{i=1}^n (\cosh^2(\lambda r_i) - 1)} \\
&\geq \frac{\lambda}{\sqrt{n}}\sum_{i=1}^n \sqrt{\cosh^2(\lambda r_i) - 1} \geq \frac{\lambda}{\sqrt{n}}\sum_{i=1}^n (\cosh(\lambda r_i) - 1) \\
&= \frac{\lambda}{\sqrt{n}}(\Phi(r) - n).
\end{aligned}$$

For the last claim, using $\sinh(r+\delta) = \sinh r\cosh\delta + \cosh r \sinh\delta$ and $\cosh r \leq |\sinh r| + 1$, for $|\delta| \leq \frac{1}{5}$, we have

$$\begin{aligned}
|\sinh(r+\delta) - \sinh(r)| &\leq |\cosh\delta - 1|\cdot|\sinh r| + |\sinh\delta|\cdot\cosh r \\
&\leq (|\cosh\delta - 1| + |\sinh\delta|)\cdot|\sinh r| + |\sinh\delta| \\
&\leq \frac{1}{4}|\sinh r| + \frac{1}{4}.
\end{aligned}$$

Using that $\nabla\Phi(r) = (\lambda\sinh(\lambda r_i))_{i=1}^n$, for $\|\delta\|_\infty \leq \frac{1}{5\lambda}$, we have

$$\|\nabla\Phi(r+\delta) - \nabla\Phi(r)\|_2 \leq \frac{1}{4}\|\nabla\Phi(r)\|_2 + \frac{\sqrt{n}\lambda}{4}. \tag{3.4}$$

Since $\Phi(r) \geq 4n$, we have that $\|\nabla\Phi(r)\|_2 \geq 3\sqrt{n}\lambda$ and hence (3.4) shows that

$$\|\nabla\Phi(r+\delta) - \nabla\Phi(r)\|_2 \leq (\frac{1}{4} + \frac{1}{12})\|\nabla\Phi(r)\|_2 = \frac{1}{3}\|\nabla\Phi(r)\|_2.$$

□

We collect some basic bounds on the step in the following lemma.

Lemma 13. *Using the notation in* `RobustStep` *(Algorithm 3). Under the invariant $\Phi((xs-t)/t) \leq 16n$, we have $\|xs - t\|_\infty \leq \frac{t}{16}$, $\|\delta_x/x\|_2 \leq \frac{1}{16\lambda}$, and $\|\delta_s/s\|_2 \leq \frac{1}{16\lambda}$.*

Proof. Using $\Phi((xs-t)/t) \leq 16n$ and Lemma 12, we have

$$\|xs - t\|_\infty \leq \frac{t\log(32n)}{\lambda} \leq \frac{t}{16}$$

By Lemma 4, we have $\overline{\mathbf{X}}^{-1}\delta_x = (\mathbf{I} - \mathbf{P})(\overline{\delta}_\mu/\overline{\mu})$ where $\overline{\mu} = \overline{xs}$ and $\mathbf{P} = \overline{\mathbf{S}}^{-1}\mathbf{A}^\top(\mathbf{A}\overline{\mathbf{S}}^{-1}\overline{\mathbf{X}}\mathbf{A}^\top)^{-1}\mathbf{A}\overline{\mathbf{X}}$. By Lemma 5, we have

$$\|\delta_x/\overline{x}\|_{\overline{\mu}} = \|(\mathbf{I} - \mathbf{P})(\overline{\delta}_\mu/\overline{\mu})\|_{\overline{\mu}} \leq \|\overline{\delta}_\mu/\overline{\mu}\|_{\overline{\mu}}.$$

Using that $\|xs-t\|_\infty \le \frac{t}{16}$, $\|\ln\overline{x}-\ln x\|_\infty \le \frac{1}{48}$, $\|\ln\overline{s}-\ln s\|_\infty \le \frac{1}{48}$, we have $\overline{\mu} \ge \frac{10}{11}t$ and hence

$$\|\delta_x/x\|_2 \le e^{1/48}\sqrt{\frac{11}{10t}}\|\delta_x/\overline{x}\|_{\overline{\mu}} \le e^{1/48}\sqrt{\frac{11}{10t}}\|\overline{\delta}_\mu\|_{\overline{\mu}^{-1}} \le e^{1/48}\frac{11}{10t}\|\overline{\delta}_\mu\|_2$$

Using the formula $\overline{\delta}_\mu = -\frac{t'}{32\lambda}\frac{\overline{g}}{\|\overline{g}\|_2}$, we have

$$\|\delta_x/x\|_2 \le e^{1/48}\frac{11}{10}\frac{t'}{32\lambda t} \le \frac{1}{16\lambda}.$$

The same proof gives $\|\delta_s/s\|_2 \le \frac{1}{16\lambda}$. $\square$

Using this, we prove the algorithm `RobustStep` satisfies the invariant on the distance.

Lemma 14. *Suppose that the input satisfies $(x,s) \in \mathcal{P}^\circ \times \mathcal{D}^\circ$ and $\Phi((xs-t_{\text{start}})/t_{\text{start}}) \le 16n$. Let $x^{(k)}, s^{(k)}, t^{(k)}$ be the x,s,t computed in the* `RobustStep` *after the k-th step. Let $\Phi^{(k)} = \Phi((x^{(k)}s^{(k)}-t^{(k)})/t^{(k)})$. Then, we have*

$$\Phi^{(k+1)} \le \begin{cases} 12n & \text{if } \Phi^{(k)} \le 8n \\ \Phi^{(k)} & \text{otherwise} \end{cases}.$$

Furthermore, we have that $\|r^{(k+1)}-r^{(k)}\|_2 \le \frac{1}{16\lambda}$ where $r^{(k)} = (x^{(k)}s^{(k)}-t^{(k)})/t^{(k)}$.

Proof. Fix some iteration k. Let $x = x^{(k)}, s = s^{(k)}, t = t^{(k)}, x' = x^{(k+1)}, s' = s^{(k+1)}$ and $t' = t^{(k+1)}$. We define $r = (xs-t)/t$ and $r' = (x's'-t')/t'$. By the definition of δ_x and δ_s, we have $\overline{\mathbf{S}}\delta_x + \overline{\mathbf{X}}\delta_s = \overline{\delta}_\mu = -\frac{t'}{32\lambda}\frac{\overline{g}}{\|\overline{g}\|_2}$ and hence

$$\begin{aligned}\frac{x's'-t'}{t'} &= \frac{(x+\delta_x)(s+\delta_s)-t'}{t'} = \frac{xs+s\delta_x+x\delta_s+\delta_x\delta_s-t'}{t'}\\ &= \frac{xs-t'+\overline{s}\delta_x+\overline{x}\delta_s+(s-\overline{s})\delta_x+(x-\overline{x})\delta_s+\delta_x\delta_s}{t'}\\ &= \frac{xs-t}{t} - \frac{1}{32\lambda}\cdot\frac{\overline{g}}{\|\overline{g}\|_2} + \eta \end{aligned} \tag{3.5}$$

where the error term

$$\eta = (\frac{t}{t'}-1)\frac{xs}{t} + \frac{(s-\overline{s})\delta_x+(x-\overline{x})\delta_s+\delta_x\delta_s}{t'}.$$

Now, we bound the error term η. Using Lemma 13 ($\|\delta_x/x\|_2 \le \frac{1}{16\lambda}$, $\|\delta_s/s\|_2 \le \frac{1}{16\lambda}$, $\|xs-t\|_\infty \le \frac{t}{16}$) and the definition of the algorithm ($\lambda \ge 16$, $|t'-t| \le \frac{t'}{128\lambda\sqrt{n}} \le \frac{t}{128\lambda\sqrt{n}}$, $\|\ln\overline{x}-\ln x\|_\infty \le \frac{1}{48}$, $\|\ln\overline{s}-\ln s\|_\infty \le \frac{1}{48}$), we have

$$\begin{aligned}\|\eta\|_2 \le& |\frac{t}{t'}-1|\|\frac{xs}{t}\|_\infty\sqrt{n} + \|\frac{xs}{t'}\|_\infty\|\frac{s-\overline{s}}{s}\|_\infty\|\frac{\delta_x}{x}\|_2\\ &+ \|\frac{xs}{t'}\|_\infty\|\frac{x-\overline{x}}{x}\|_\infty\|\frac{\delta_s}{s}\|_2 + \|\frac{xs}{t'}\|_\infty\|\frac{\delta_x}{x}\|_2\|\frac{\delta_s}{s}\|_2\\ \le& \frac{1}{128\lambda}\frac{17}{16} + \frac{9}{8}(e^{1/48}-1)(\frac{1}{16\lambda}+\frac{1}{16\lambda}) + \frac{9}{8}(\frac{1}{16\lambda}\frac{1}{16\lambda}) \le \frac{1}{80\lambda}.\end{aligned} \tag{3.6}$$

In particular, we use (3.5) and (3.6) to get

$$\|r-r'\|_2 \le \frac{1}{32\lambda} + \|\eta\|_2 \le \frac{1}{16\lambda}.$$

This proves the conclusion about r.

Case 1: $\Phi(r) \le 8n$.

The definition of Φ together with the fact $\|r-r'\|_2 \le \frac{1}{16\lambda}$ implies that $\Phi(r') \le \frac{3}{2}\Phi(r) \le 12n$.

Case 2: $\Phi(r) \ge 8n$.

Mean value theorem shows there is $\widetilde{r}$ between r and r' such that

$$\Phi(r') = \Phi(r) + \langle\nabla\Phi(\widetilde{r}), r'-r\rangle = \Phi(r) + \left\langle\nabla\Phi(\widetilde{r}), -\frac{1}{32\lambda}\frac{\overline{g}}{\|\overline{g}\|_2}+\eta\right\rangle$$

where we used (3.5) at the end. Using $\|r-r'\|_2 \leq \frac{1}{16\lambda}$ and $\|\overline{r}-r\|_\infty \leq \frac{1}{48\lambda}$ (by assumption), we have $\|\overline{r}-\widetilde{r}\|_\infty \leq \frac{1}{5\lambda}$. Since $\Phi(r) \geq 8n$, we have $\Phi(\overline{r}) \geq 4n$ and hence Lemma 12 shows that

$$\|\nabla\Phi(\widetilde{r}) - \nabla\Phi(\overline{r})\|_2 \leq \frac{1}{3}\|\nabla\Phi(\overline{r})\|_2.$$

Using $\overline{g} = \nabla\Phi(\overline{r})$ and letting $\eta_2 = \nabla\Phi(\widetilde{r}) - \nabla\Phi(\overline{r})$, we have

$$\Phi(r') - \Phi(r) = \left\langle \overline{g} + \eta_2, -\frac{1}{32\lambda}\frac{\overline{g}}{\|\overline{g}\|_2} + \eta \right\rangle = -\frac{1}{32\lambda}\|\overline{g}\|_2 - \frac{1}{32\lambda}\eta_2^\top \frac{\overline{g}}{\|\overline{g}\|_2} + \overline{g}^\top\eta + \eta_2^\top\eta.$$

Using $\|\eta_2\|_2 \leq \frac{1}{3}\|\overline{g}\|_2$ and $\|\eta\|_2 \leq \frac{1}{80\lambda}$ (3.6), we have

$$\Phi(r') - \Phi(r) \leq -\frac{1}{32\lambda}\|\overline{g}\|_2 + \frac{1}{32\lambda}\cdot\frac{1}{3}\|\overline{g}\|_2 + \|\overline{g}\|_2\cdot\frac{1}{80\lambda} + \frac{1}{3}\|\overline{g}\|_2\cdot\frac{1}{80\lambda} \leq -\frac{1}{240\lambda}\|\overline{g}\|_2$$

Using Lemma 12, we have $\|\overline{g}\|_2 \geq \frac{\lambda}{\sqrt{n}}(\Phi(\overline{r}) - n) \geq 3\lambda\sqrt{n}$. Hence, we have

$$\Phi(r') \leq \Phi(r) - \frac{\sqrt{n}}{80} < 16n. \tag{3.7}$$

The potential actually decreases in this case. This proves the conclusion about Φ. □

Exercise 15. Show that we can use the potential function $\Phi = \sum_{i=1}^n \exp(\lambda|v_i|)$ to get a similar conclusion.

3.2 Selecting $\overline{x}, \overline{s}$ and $\overline{r}$

Each step of the robust interior point method solves the linear system

$$\begin{pmatrix} \overline{\mathbf{S}} & \overline{\mathbf{X}} & \mathbf{0} \\ \mathbf{A} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{A}^\top \end{pmatrix} \begin{pmatrix} \delta_x \\ \delta_s \\ \delta_y \end{pmatrix} = \begin{pmatrix} \nabla\Phi(\overline{r}) \\ 0 \\ 0 \end{pmatrix}$$

for some vectors $\overline{x}, \overline{s}, \overline{r}$ such that $\|\ln\overline{x} - \ln x\|_\infty \leq \frac{1}{48}$, $\|\ln\overline{s} - \ln s\|_\infty \leq \frac{1}{48}$, $\|\overline{r} - r\|_\infty \leq \frac{1}{48\lambda}$. The key observation is that only a few coordinates of x, s and r change significantly each step and hence we can maintain the solution of the linear system instead of computing from scratch. In this section, we discuss how to select $\overline{x}, \overline{s}, \overline{r}$ with as few updates as possible while maintaining the invariants.

First, we observe that $\ln x, \ln s$ and r change by $O(1)$ in ℓ_2 norm in each step.

Lemma 16. *Define $x^{(k)}, s^{(k)}, r^{(k)}$ according to Lemma 14. Then, $\|\ln x^{(k+1)} - \ln x^{(k)}\|_2$, $\|\ln s^{(k+1)} - \ln s^{(k)}\|_2$ and $\|r^{(k+1)} - r^{(k)}\|_2$ are all bounded by $1/(8\lambda)$.*

Proof. Lemma 5 shows that

$$\|(x^{(k+1)} - x^{(k)})/x^{(k)}\|_{\mu^{(k)}} \leq \|\overline{\delta}_\mu/\mu^{(k)}\|_{\mu^{(k)}} \tag{3.8}$$

where $\mu^{(k)} = \overline{x}^{(k)}\overline{s}^{(k)}$ and $\overline{x}^{(k)}, \overline{s}^{(k)}$ are the $\overline{x}, \overline{s}$ used in the k-th step.

To bound $\mu^{(k)}$, Lemma 14 shows that the invariant $\Phi(r^{(k)}) \leq 16n$ holds and hence Lemma 12 shows that (recall $\lambda = 16\log(40n)$):

$$\|(x^{(k)}s^{(k)} - t^{(k)})/t^{(k)}\|_\infty = \|r^{(k)}\|_\infty \leq \frac{\log(32n)}{\lambda} \leq \frac{1}{16}.$$

Together with the fact that $\|\ln\overline{x}^{(k)} - \ln x^{(k)}\|_\infty \leq \frac{1}{48}$, $\|\ln\overline{s}^{(k)} - \ln s^{(k)}\|_\infty \leq \frac{1}{48}$, we have

$$\|\frac{\overline{x}^{(k)}\overline{s}^{(k)} - t^{(k)}}{t^{(k)}}\|_\infty \leq \frac{1}{8}.$$

Using this on (3.8) gives $\|(x^{(k+1)} - x^{(k)})/x^{(k)}\|_2 \leq \frac{8}{7}\frac{1}{t^{(k)}}\|\overline{\delta}_\mu\|_2$.Using $\overline{\delta}_\mu = -\frac{t^{(k+1)}}{32\lambda}\frac{\overline{g}}{\|\overline{g}\|_2}$, we have

$$\|(x^{(k+1)} - x^{(k)})/x^{(k)}\|_2 \leq \frac{1}{28\lambda}.$$

To translate the bound to log scale, we note that $|\ln(1+t)| \leq 2|t|$ for all $|t| \leq \frac{1}{2}$ and hence

$$\|\ln x^{(k+1)} - \ln x^{(k)}\|_2 = \|\ln(1 + \frac{x^{(k+1)} - x^{(k)}}{x^{(k)}})\|_2 \leq \frac{1}{14\lambda}.$$

The bound for $\|\ln s^{(k+1)} - \ln s^{(k)}\|_2$ is similar.

The bound for $\|r^{(k+1)} - r^{(k)}\|_2$ follows from Lemma 14. □

Now the question is how to select $\ln \overline{x}, \ln \overline{s}$ and $\overline{r}$ such that they are close to $\ln x, \ln s$ and r in ℓ_∞ norm. If the cost of updating the inverse of a matrix is linear in the rank of the update, then we can simply update any coordinate of $\overline{x}, \overline{s}$ and $\overline{r}$ whenever they violate the condition. However, due to fast matrix multiplication, the average cost (per rank) of update is lower when the rank of update is large. Therefore, it is beneficial to update coordinates preemptively.

Now we state the algorithm for selecting $\overline{x}, \overline{s}$ and $\overline{r}$. This algorithm is a general algorithm for maintaining a vector $\overline{v}$ such that $\|\overline{v} - v\|_\infty \leq \delta$. For every 2^ℓ steps, the algorithm updates the coordinate of the vector $\overline{v}$ if that coordinate has changed by more than $\delta/(2\log n)$ between this step and 2^ℓ steps earlier.

Algorithm 4: `SelectVector`$(\overline{v}, v^{(0)}, v^{(1)}, \cdots, v^{(k)}, \delta)$

Let $I = \{\}$ be the set of coordinates to update.
for $\ell = 0, 1, \cdots, \lceil \log n \rceil$ **do**
  if $k = 0 \mod 2^\ell$ **then**
    if $\ell = \lceil \log n \rceil$ **then**
      $I = [n]$.
    else
      $I = I \cup \{i : |v_i^{(k)} - v_i^{(k-2^\ell)}| \geq \delta/(2\lceil \log n \rceil)\}$.
    end
  end
end
$\overline{v}_i \leftarrow v_i^{(k)}$ for all $i \in I$
Return $\overline{v}$

Lemma 17. *Given vectors $v^{(0)}, v^{(1)}, v^{(2)}, \cdots$ arriving in a stream, suppose that $\|v^{(k+1)} - v^{(k)}\|_2 \leq \beta$ for all k. For any $\frac{1}{2} > \delta > 0$, define the vector $\overline{v}^{(0)} = v^{(0)}$ and $\overline{v}^{(k)} =$ `SelectVector`$(\overline{v}^{(k-1)}, v^{(0)}, v^{(1)}, \cdots, v^{(k)}, \delta)$. Then, we have that*

- $\|\overline{v}^{(k)} - v^{(k)}\|_\infty \leq \delta$ *for all k.*
- $\|\overline{v}^{(k)} - \overline{v}^{(k-1)}\|_0 \leq O(\min(2^{2\ell_k}(\beta/\delta)^2 \log^2 n, n))$ *where ℓ_k is the largest integer ℓ with $k = 0 \mod 2^\ell$.*

Proof. For bounding the error, we first fix some coordinate $i \in [n]$. Let k' be the iteration when $\overline{v}_i$ was last updated, namely, $\overline{v}_i^{(k)} = \overline{v}_i^{(k')} = v_i^{(k')}$. Since we set $\overline{v} \leftarrow v$ every $2^{\lceil \log n \rceil}$ steps, we have $k - 2^{\lceil \log n \rceil} \leq k' < k$. We can write $k' = k_0 < k_1 < k_2 < \cdots < k_s = k$ such that $k_{i+1} - k_i$ is a power of 2 and $k_{i+1} - k_i$ divides k_{i+1} with $s \leq 2\lceil \log n \rceil$. Hence, we have that

$$v_i^{(k)} - \overline{v}_i^{(k)} = v_i^{(k_s)} - v_i^{(k_0)} = \sum_{j=0}^{s-1} (v_i^{(k_{j+1})} - v_i^{(k_j)}).$$

Since $\overline{v}_i$ is not updated since step k', we have $|v_i^{(k_{j+1})} - v_i^{(k_j)}| \leq \delta/(2\lceil \log n \rceil)$ and hence $|v_i^{(k)} - \overline{v}_i^{(k)}| \leq \delta$. Since this holds for every i, we have that $\|\overline{v}^{(k)} - v^{(k)}\|_\infty \leq \delta$.

For the sparsity of $\overline{v}^{(k+1)} - \overline{v}^{(k)}$, we first bound the size of the set $I_\ell \stackrel{\text{def}}{=} \{i : |v_i^{(k)} - v_i^{(k-2^\ell)}| \geq \delta/(2\lceil \log n \rceil)\}$. Note that

$$|I_\ell| \cdot \frac{\delta^2}{5\log^2 n} \leq \sum_{i=1}^n |v_i^{(k)} - v_i^{(k-2^\ell)}|^2 \leq 2^\ell \sum_{i=1}^n \sum_{t=k-2^\ell}^{k-1} |v_i^{(t+1)} - v_i^{(t)}|^2 \leq 2^{2\ell}\beta^2$$

where we used $\|v^{(t+1)} - v^{(t)}\|_2 \leq \beta$ at the end. Hence, we have $|I_\ell| = O(2^{2\ell}(\beta/\delta)^2 \log^2 n)$. Hence, the total number of changes is bounded by

$$|I| \leq \sum_{\ell=0}^{\ell_k} |I_\ell| = O(2^{2\ell_k}(\beta/\delta)^2 \log^2 n).$$

□

3.3 Framework

In this section, we give the meta theorem by combining the algorithm `RobustStep` and `SelectVector`. The algorithm is given in Algorithm 5.

Algorithm 5: $\texttt{FastRobustStep}(\mathbf{A}, x, s, t_{\text{start}}, t_{\text{end}})$

Define $\Phi(r)$ according to (3.2) with $\lambda = 16\log(40n)$.
Invariant: $(x, s) \in \mathcal{P}^{\circ} \times \mathcal{D}^{\circ}$ and $\Phi((xs - t)/t) \leq 16n$.
Let $h = 1/(128\lambda\sqrt{n})$, and let n be the number of columns in $\mathbf{A}$.
Set $k \leftarrow 0$, $t^{(0)} = t_{\text{start}}$, $x^{(0)} = \overline{x}^{(0)} = x$, $s^{(0)} = \overline{s}^{(0)} = s$, $r^{(0)} = \overline{r}^{(0)} = \frac{xs - t^{(0)}}{t^{(0)}}$.
repeat
  Let $t^{(k+1)} = \max(t^{(k)}/(1+h), t_{\text{end}})$, $\overline{\delta}_{\mu} = -\frac{t^{(k+1)}}{32\lambda}\frac{\overline{g}}{\|\overline{g}\|_2}$, $\overline{g} = \nabla\Phi(\overline{r}^{(k)})$.
  Find δ_x, δ_s such that

$$\begin{aligned} \overline{\mathbf{S}}^{(k)}\delta_x + \overline{\mathbf{X}}^{(k)}\delta_s &= \overline{\delta}_{\mu}, \\ \mathbf{A}\delta_x &= 0, \\ \mathbf{A}^{\top}\delta_y + \delta_s &= 0. \end{aligned} \tag{3.9}$$

  Set $x^{(k+1)} \leftarrow x^{(k)} + \delta_x$, $s^{(k+1)} \leftarrow s^{(k)} + \delta_s$, $r^{(k+1)} \leftarrow (x^{(k+1)}s^{(k+1)} - t^{(k+1)})/t^{(k+1)}$.
  $\ln \overline{x}^{(k+1)} = \texttt{SelectVector}(\ln \overline{x}^{(k)}, \ln x^{(0)}, \ln x^{(1)}, \cdots, \ln x^{(k+1)}, 1/48)$.
  $\ln \overline{s}^{(k+1)} = \texttt{SelectVector}(\ln \overline{s}^{(k)}, \ln s^{(0)}, \ln s^{(1)}, \cdots, \ln s^{(k+1)}, 1/48)$.
  $\overline{r}^{(k+1)} = \texttt{SelectVector}(\overline{r}^{(k)}, r^{(0)}, r^{(1)}, \cdots, r^{(k+1)}, 1/(48\lambda))$.
  Set $k \leftarrow k+1$.
until $t^{(k)} = t_{\text{end}}$;
Return $(x^{(k)}, s^{(k)})$.

The intermediate variables $x^{(k)}, s^{(k)}, \overline{x}^{(k)}, \overline{s}^{(k)}, \overline{r}^{(k)}$ can be stored implicitly and the cost of $\texttt{SelectVector}$ roughly depends on how many coordinates of $\overline{x}, \overline{s}, \overline{r}$ are updated. Hence, the bottleneck of $\texttt{FastRobustStep}$ is solving the linear systems in 3.9. The cost of solving a sequence of linear systems in general depends on the number of entries changed. Combining the lemmas proved in the previous two subsections, we can prove the correctness of $\texttt{FastRobustStep}$ and bound the number of entries changed.

Lemma 18. *Suppose that the input satisfies $(x, s) \in \mathcal{P}^{\circ} \times \mathcal{D}^{\circ}$ and $\Phi_{\text{start}} \stackrel{\text{def}}{=} \Phi((xs - t_{\text{start}})/t_{\text{start}}) \leq 16n$. After $O(\sqrt{n}\log(t_{\text{start}}/t_{\text{end}}))$ iterations, $\texttt{FastRobustStep}$ outputs (x, s) such that*

$$\Phi((xs - t_{\text{end}})/t_{\text{end}}) \leq \max(\Phi_{\text{start}}, 12n).$$

Furthermore, in the k-th step, we have

$$\|\overline{x}^{(k)} - \overline{x}^{(k-1)}\|_0 + \|\overline{s}^{(k)} - \overline{s}^{(k-1)}\|_0 + \|\overline{r}^{(k)} - \overline{r}^{(k-1)}\|_0 \leq O(\min(2^{2\ell_k}\log^2 n, n))$$

where ℓ_k is the largest integer ℓ with $k = 0 \mod 2^{\ell}$.

Proof. Note that $\texttt{FastRobustStep}$ is an instantiation of $\texttt{RobustStep}$, and its output is analyzed in Lemma 14. The sparsity guarantees for $\overline{x}$, $\overline{s}$, and $\overline{r}$ follow from Lemma 17 with $\beta = \frac{1}{8\lambda}$ (Lemma 16) and $\delta \geq \frac{1}{48\lambda}$. □

As we see from Section 2.4, we can solve a linear program by calling $\texttt{FastRobustStep}$ (or $\texttt{L2Step}$) twice and that

$$\frac{t_{\text{start}}}{t_{\text{end}}} = \left(\frac{nR}{\delta r}\right)^{O(1)}$$

for both calls. Hence, the total number of steps is bounded by

$$T = O(\sqrt{n}\log(\frac{nR}{\delta r}))$$

and the number of entries changed is bounded by

$$\sum_{k=1}^{T} O(\min(2^{2\ell_k}\log^2 n, n)) = O(n\log n \cdot \log(\frac{nR}{\delta r})).$$

Note that we need to change at least n entries to get an approximate solution. Hence, the number of changes is optimal up to a polylogarithmic factor.

3.4 Inverse Maintenance

In this section, we discuss how to maintain the solution of the linear system (Newton step) efficiently. There are many different ways of maintaining the solution depending on the structure of $\mathbf{A}$. In this note, we consider the generic case that $\mathbf{A}$ is an arbitrary matrix. Although both the matrix and the vector of the Newton step change during the algorithm, we can simplify by moving the vector into the matrix.

Fact 19. *For any invertible matrix $\mathbf{M} \in \mathbb{R}^{n\times n}$ and any vector $v \in \mathbb{R}^n$, we have*

$$\begin{bmatrix} \mathbf{M} & v \\ 0 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{M}^{-1} & \mathbf{M}^{-1}v \\ 0 & -1 \end{bmatrix}.$$

Hence, the question of maintaining the solution reduces to the problem of maintaining a column of the inverse of the matrix

$$\mathbf{M}_{\overline{x},\overline{s},\overline{r}} \stackrel{\text{def}}{=} \begin{pmatrix} \overline{\mathbf{S}} & \overline{\mathbf{X}} & \mathbf{0} & \nabla\Phi(\overline{r}) \\ \mathbf{A} & \mathbf{0} & \mathbf{0} & 0 \\ \mathbf{0} & \mathbf{I} & \mathbf{A}^\top & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{3.10}$$

When we update $\overline{x},\overline{s},\overline{r}$ to $\overline{x}+\delta_{\overline{x}},\overline{s}+\delta_{\overline{s}},\overline{r}+\delta_{\overline{r}}$ with q coordinates modified in total, there are only q entries in $\mathbf{M}$ that change. Hence, we can compute the update of the inverse of $\mathbf{M}$ using the Woodbury matrix identity (Equation (3.11)). The idea of using the Woodbury identity together with fast matrix multiplication goes back to Vaidya [18].

Lemma 20. *Given vectors $\overline{x},\overline{s},\overline{r} \in \mathbb{R}^n$ and an update $\delta_{\overline{x}},\delta_{\overline{s}},\delta_{\overline{r}} \in \mathbb{R}^n$, let $q = \|\delta_{\overline{x}}\|_0 + \|\delta_{\overline{s}}\|_0 + \|\delta_{\overline{r}}\|_0$ and $T_{m,n,\ell}$ be the cost of multiplying an $m\times n$ matrix with an $n\times \ell$ matrix. Then,*

- *Given $\mathbf{M}^{-1}_{\overline{x},\overline{s},\overline{r}}$, we can compute $\mathbf{M}^{-1}_{\overline{x}+\delta_{\overline{x}},\overline{s}+\delta_{\overline{s}},\overline{r}+\delta_{\overline{r}}}$ in time $O(T_{n,q,n})$.*
- *Given $\mathbf{M}^{-1}_{\overline{x},\overline{s},\overline{r}}$ and $\mathbf{M}^{-1}_{\overline{x},\overline{s},\overline{r}}b$, we can compute $\mathbf{M}^{-1}_{\overline{x}+\delta_{\overline{x}},\overline{s}+\delta_{\overline{s}},\overline{r}+\delta_{\overline{r}}}b$ in time $O(T_{q,q,q}+nq)$.*

Proof. We write $\mathbf{M}_0 = \mathbf{M}_{\overline{x},\overline{s},\overline{r}}$ and $\mathbf{M}_1 = \mathbf{M}_{\overline{x}+\delta_{\overline{x}},\overline{s}+\delta_{\overline{s}},\overline{r}+\delta_{\overline{r}}}$. Note that $\mathbf{M}_1$ and $\mathbf{M}_0$ are off by just q entries. Hence, we can write

$$\mathbf{M}_1 = \mathbf{M}_0 + \mathbf{UCV}$$

where $\mathbf{U}$ consists of q columns of the identity matrix, $\mathbf{C}$ is a $q\times q$ matrix and $\mathbf{V}$ consists of q rows of the identity matrix. Hence, the Woodbury matrix identity shows that

$$\mathbf{M}_1^{-1} = (\mathbf{M}_0 + \mathbf{UCV})^{-1} = \mathbf{M}_0^{-1} - \mathbf{M}_0^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}\mathbf{M}_0^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{M}_0^{-1}. \tag{3.11}$$

Note that $\mathbf{M}_0^{-1}\mathbf{U}$, $\mathbf{V}\mathbf{M}_0^{-1}\mathbf{U}$, $\mathbf{V}\mathbf{M}_0^{-1}$ are just blocks of $\mathbf{M}_0^{-1}$ and no computation is needed. Hence, we can compute $(\mathbf{C}^{-1} + \mathbf{V}\mathbf{M}_0^{-1}\mathbf{U})^{-1}$ in the time to invert two $q\times q$ matrices, which is $O(T_{q,q,q})$. The rest of the formula can be computed in $O(T_{n,q,q} + T_{n,q,n}) = O(T_{n,q,n})$ time. In total, the runtime is $O(T_{n,q,n})$.

For computing $\mathbf{M}_1^{-1}b$, we note that

$$\mathbf{M}_1^{-1}b = \mathbf{M}_0^{-1}b - \mathbf{M}_0^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}\mathbf{M}_0^{-1}\mathbf{U})^{-1}\mathbf{V}\mathbf{M}_0^{-1}b.$$

Since $\mathbf{M}_0^{-1}b$ is given, the above formula can be computed in $O(T_{q,q,q}+nq)$ time where the $O(nq)$ term comes from multiplying a $q\times n$ matrix with an n-vector and a $n\times q$ matrix with a q-vector. □

To use the previous lemma, we use the following estimate for $T_{n,r,n}$.

Definition 21. The exponent of matrix multiplication ω is the infimum among all $\omega \geq 0$ such that it takes $n^{\omega+o(1)}$ time to multiply an $n\times n$ matrix by an $n\times n$ matrix. The dual exponent of matrix multiplication α is the supremum among all $\alpha \geq 0$ such that it takes $n^{2+o(1)}$ time to multiply an $n\times n$ matrix by an $n\times n^\alpha$ matrix. Currently, $\omega \leq 2.3729$ [3, 24, 11, 1] and $\alpha \geq 0.3138$ [11, 5].

Lemma 22. *For $r \leq n$, we have $T_{n,r,n} = n^{2+o(1)} + n^{\omega-\frac{\omega-2}{1-\alpha}+o(1)}r^{\frac{\omega-2}{1-\alpha}}$.*

Algorithm 6: `InverseMaintenance()`

Set $k \leftarrow 0$ and ℓ_* such that $2^{2\ell_*} = \Theta(\min(n^\alpha, n^{2/3}))$.
while *Not finished* **do**
  Obtain the current $\overline{x}^{(k)}, \overline{s}^{(k)}, \overline{r}^{(k)}$.
  if $k = 0 \mod 2^{\ell_*}$ **then**
    Update $\mathbf{T}$ to $\mathbf{M}^{-1}_{\overline{x}^{(k)},\overline{s}^{(k)},\overline{r}^{(k)}}$ using Lemma 20.
    $u \leftarrow \mathbf{T}e_{2n+d+1}$, $v \leftarrow u$.
  else
    Update v to $\mathbf{M}^{-1}_{\overline{x}^{(k)},\overline{s}^{(k)},\overline{r}^{(k)}}e_{2n+d+1}$ using vector u and Lemma 20.
  end
  $k \leftarrow k + 1$
  Let (δ_x, δ_s) be the first $2n$ coordinates of v.
  Yield (δ_x, δ_s)
end

We note that in the algorithm above, we do not compute the inverse of $\mathbf{M}$ in every iteration, which would be too expensive. Rather, we compute $\mathbf{M}^{-1}b$ as needed by using the previously computed $\mathbf{M}^{-1}$ (from possibly many iterations ago) with a low-rank update using the Woodbury formula that we maintain.

Combining Lemma 20 with Lemma 17, we have the following guarantee.

Lemma 23. *Using* `InverseMaintenance` *to implement Line 3.9 of* `FastRobustStep` *(Algorithm 5),* `FastRobustStep` *takes in total time*

$$O((n^{\omega+o(1)} + n^{2+1/6+o(1)} + n^{5/2-\alpha/2+o(1)}) \log(t_{\text{start}}/t_{\text{end}})).$$

Proof. The bottleneck of `FastRobustStep` is the time to update v and $\mathbf{T}$. This depends on the number of coordinates updated in $\overline{x}, \overline{s}, \overline{r}$. Lemma 16 shows that $\ln x, \ln s$ and r change by at most $\alpha \stackrel{\text{def}}{=} 1/(8\lambda)$ in ℓ_2 norm per step. Since we set the error of `SelectVector` to be $\delta \stackrel{\text{def}}{=} 1/(48\lambda)$ (or larger), Lemma 17 shows that $q_k \stackrel{\text{def}}{=} O(2^{2\ell_k} \log^2 n)$ coordinates in $\overline{x}, \overline{s}, \overline{r}$ are updated at the k-th step where ℓ_k is the largest integer ℓ with $k = 0 \mod 2^\ell$. We can now bound all the computation costs as follows.

Cost of updating v: We update u whenever $k = 0 \mod 2^{\ell_*}$. Within that 2^{ℓ_*} steps, the number of coordinates updated in $\overline{x}, \overline{s}, \overline{r}$ is bounded by

$$q \stackrel{\text{def}}{=} \sum_{k=1}^{2^{\ell_*}-1} q_k = \sum_{k=1}^{2^{\ell_*}-1} O(2^{2\ell_k} \log^2 n) = O(2^{2\ell_*} \log^2 n).$$

Therefore, $\mathbf{M}_{\overline{x}^{(k)},\overline{s}^{(k)},\overline{r}^{(k)}}$ and $\mathbf{T}^{-1}$ are off by at most q coordinates. Lemma 20 shows that it takes

$$O(T_{q,q,q} + nq) = \widetilde{\widetilde{O}}(2^{2\ell_*\omega} + n2^{2\ell_*})$$

time to compute $v = \mathbf{M}^{-1}_{\overline{x}^{(k)},\overline{s}^{(k)},\overline{r}^{(k)}}e_{2n+d+1}$ using $u = \mathbf{T}e_{2n+d+1}$, where we used $\widetilde{\widetilde{O}}$ to omit $n^{o(1)}$ terms.

Cost of updating $\mathbf{T}$: For the k-th step that updates $\mathbf{T}$, the number of coordinates updated in $\overline{x}, \overline{s}, \overline{r}$ is bounded by $q + O(2^{2\ell_k} \log^2 n) = O(2^{2\ell_k} \log^2 n)$ where the first term is due to the delayed updates and the second term is due to the updates at that step. Lemma 20 shows that it takes $\widetilde{\widetilde{O}}(T_{n,n,2^{2\ell_k}})$ time to update $\mathbf{T}$. Since $2^{2\ell_k}$ updates happen every 2^{ℓ_k} iterations, the amortized cost is

$$\begin{aligned}
\widetilde{\widetilde{O}}\Big(\sum_{\ell=\ell_*}^{\frac{1}{2}\log n} 2^{-\ell} T_{n,n,2^{2\ell}}\Big) &= \widetilde{\widetilde{O}}\Big(\sum_{\ell=\ell_*}^{\frac{1}{2}\log n} (n^{\omega - \frac{\omega-2}{1-\alpha}} 2^{2\ell \cdot \frac{\omega-2}{1-\alpha} - \ell} + n^2 2^{-\ell})\Big) \\
&= \widetilde{\widetilde{O}}\Big(\sum_{\ell=\ell_*}^{\frac{1}{2}\log n} n^{\omega - \frac{\omega-2}{1-\alpha}} 2^{2\ell \cdot \frac{\omega-2}{1-\alpha} - \ell} + n^2 2^{-\ell_*}\Big)
\end{aligned}$$

where we used Lemma 22. The sum above is dominated by either the term at $\ell = \ell_*$ or the term at $\ell = \frac{1}{2}\log n$. Hence, the amortized cost of updating $\mathbf{T}$ is

$$\widetilde{\widetilde{O}}(n^{\omega - \frac{\omega-2}{1-\alpha}} 2^{2\ell_* \cdot \frac{\omega-2}{1-\alpha} - \ell_*} + n^{\omega - \frac{1}{2}} + n^2 2^{-\ell_*}) = \widetilde{\widetilde{O}}(n^{\omega - \frac{1}{2}} + n^2 2^{-\ell_*})$$

where we used $n^{\omega-\frac{\omega-2}{1-\alpha}}2^{2\ell_*\cdot\frac{\omega-2}{1-\alpha}} \leq n^2$ since $2^{2\ell_*} \leq n^\alpha$.

Cost of initializing $\mathbf{T}$ and u: $\widetilde{\widetilde{O}}(n^\omega)$.

Since there are $\sqrt{n}\log(t_{\text{start}}/t_{\text{end}})$ steps up to constants, the total cost is

$$\begin{aligned}&\widetilde{\widetilde{O}}(n^\omega + \sqrt{n}\log(t_{\text{start}}/t_{\text{end}})(2^{2\ell_*\omega} + n2^{2\ell_*} + n^{\omega-\frac{1}{2}} + n^2 2^{-\ell_*}))\\ =&\widetilde{\widetilde{O}}(\sqrt{n}\log(t_{\text{start}}/t_{\text{end}})(n2^{2\ell_*} + n^{\omega-\frac{1}{2}} + n^2 2^{-\ell_*}))\end{aligned}$$

where we used $2^{2\ell_*\omega} \leq n^{\alpha\omega} \leq n$. Putting $2^{2\ell_*} = \min(n^\alpha, n^{2/3})$, we have

$$\widetilde{\widetilde{O}}((n^\omega + n^{2+1/6} + n^{5/2-\alpha/2})\log(t_{\text{start}}/t_{\text{end}})).$$

□

Following Section 2.4, we can find the initial point by modifying the linear program and this gives the following theorem.

Theorem 24. *Consider a linear program $\min_{\mathbf{A}x=b,x\geq 0} c^\top x$ with n variables and d constraints. Assume the linear program has inner radius r, outer radius R and Lipschitz constant L (see Definition 8). Then we can find x such that*

$$\begin{aligned}c^\top x &\leq \min_{\mathbf{A}x=b,x\geq 0} c^\top x + \delta LR,\\ \mathbf{A}x &= b,\\ x &\geq 0.\end{aligned}$$

in time

$$O((n^{\omega+o(1)} + n^{2+1/6+o(1)} + n^{5/2-\alpha/2+o(1)})\log(R/(\delta r))).$$

If we further assume that the solution $x^* = \arg\min_{\mathbf{A}x=b,x\geq 0} c^\top x$ is unique and that $c^\top x \geq c^\top x^* + \eta LR$ for any other vertex x of $\{\mathbf{A}x = b, x \geq 0\}$ for some $\eta > \delta \geq 0$, then we have that $\|x - x^*\|_2 \leq \frac{2\delta R}{\eta}$.

Proof. The algorithm for finding x is the same as `SlowSolveLP` except that the function `L2Step` is replaced by the function `FastRobustStep`. The runtime of `FastRobustStep` is analyzed in Lemma 23. Since `FastRobustStep` is an instantiation of `RobustStep`, its output is analyzed in Lemma 14. □

Historical Note. The interior-point method was pioneered by Karmarkar [9] and developed in beautiful ways (including [17, 18, 15, 16, 12, 13]). This classical approach appeared to reach its limit of requiring the solution of $\sqrt{n}$ linear systems until the paper of Cohen, Lee and Song [2] which reduced the complexity of approximate linear programming to $n^{\omega+o(1)}\log(1/\epsilon)$, by introducing the robust central path method. Using further insights, their algorithm was derandomized by van den Brand [19]. The technique has been extended in subsequent papers to other optimization problems [14, 7, 23, 8, 20, 6], and has also played a crucial role in faster algorithms for classical combinatorial optimization problems such as matchings and flows [22, 21, 4].

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A Finding a Point on the Central Path

We continue from the discussion in Section 2.4.

First, we show that $(\overline{x}^{+}, \overline{x}^{-}, \overline{R})$ defined in Theorem 9 is indeed on the central path of the modified linear program. First, we define the dual of the modified linear program as follows

$$\mathcal{D}_{\eta} = \{(s^{+}, s^{-}, s^{\theta}) \in \mathbb{R}_{\geq 0}^{2n+1} : \mathbf{A}^{\top} y + \lambda \mathbf{1} + s^{+} = c^{+}, -\mathbf{A}^{\top} y + s^{-} = c^{-}, \lambda + s^{\theta} = 0 \text{ for some } y \in \mathbb{R}^{d} \text{ and } \lambda \in \mathbb{R}\}.$$

Lemma 25. *The modified linear program in Theorem 9 has an explicit point on the central path,$(\overline{x}^{+}, \overline{x}^{-}, \overline{R})$ at $\overline{t}$.*

Proof. Recall that we say $(x^{+}, x^{-}, x^{\theta})$ is on the central path at t if x^{+}, x^{-}, x^{θ} are positive and it satisfies the following equation

$$\begin{aligned} \mathbf{A}x^{+} - \mathbf{A}x^{-} &= b, \\ \sum_{i=1}^{n} x_i^{+} + x^{\theta} &= \widetilde{b}, \\ \mathbf{A}^{\top} y + \lambda + s^{+} &= c^{+}, \\ -\mathbf{A}^{\top} y + s^{-} &= c^{-}, \\ \lambda + s^{\theta} &= 0, \end{aligned} \tag{A.1}$$

for some $s^{+}, s^{-} \in \mathbb{R}_{>0}^{n}$, $s^{\theta} > 0$, $y \in \mathbb{R}^{d}$ and $\lambda \in \mathbb{R}$.

Now, we verify the solution $\overline{x}^{+} = \overline{t}/(c + \overline{t}/\overline{R})$, $\overline{x}^{-} = \overline{x}^{+} - \mathbf{A}^{\top}(\mathbf{A}\mathbf{A}^{\top})^{-1} b$, $\overline{x}^{\theta} = \overline{R}$, $y = 0$, $\overline{s}^{+} = \overline{t}/\overline{x}^{+}$, $\overline{s}^{-} = \overline{t}/\overline{x}^{-}$, $\overline{s}^{\theta} = \overline{t}/\overline{x}^{\theta}$, $\lambda = -\overline{s}^{\theta}$. Using $\mathbf{A}\mathbf{A}^{\top}(\mathbf{A}\mathbf{A}^{\top})^{-1} b = b$, one can check it satisfies all the equality constraints above.

For the positivity constraints, using $\|c\|_{\infty} \leq L$ and $\overline{t} \geq 4L\overline{R}$, we have

$$\frac{4}{5}\overline{R} \leq \frac{\overline{t}}{L + \overline{t}/\overline{R}} \leq \overline{x}_i^{+} \leq \frac{\overline{t}}{-L + \overline{t}/\overline{R}} \leq \frac{4}{3}\overline{R} \tag{A.2}$$

and hence $\overline{x}^{+} > 0$ and so is $\overline{s}^{+}$. Note that $x_{\circ} \stackrel{\text{def}}{=} \mathbf{A}^{\top}(\mathbf{A}\mathbf{A}^{\top})^{-1} b$ is the least square solution to $\mathbf{A}x = b$, so, we have

$$\|x_{\circ}\|_2 \leq \|x^{*}\|_2 \leq R < \frac{1}{4}\overline{R}. \tag{A.3}$$

Together with $\overline{x}_i^{+} \geq \frac{4}{5}\overline{R}$, we have

$$\overline{x}_i^{-} \geq \frac{4}{5}\overline{R} - \frac{1}{4}\overline{R} > 0$$

for all i. Hence, $\overline{x}^{-}$ and $\overline{s}^{-}$ are positive. Finally, $\overline{x}^{\theta}$ and $\overline{s}^{\theta}$ are positive. This proves that $(\overline{x}^{+}, \overline{x}^{-}, \overline{R})$ is on the central path at t. □

Next, we show that the near-central-path point (x, s) at $t = LR$ is far from the boundary $x^{+} = 0$ and is close to the boundary $x^{-} = 0$. The proof for both involves the same idea: use the optimality condition of x. Throughout the rest of the proof, we are given $(x, s) \in \mathcal{P}_{\eta} \times \mathcal{D}_{\eta}$ such that $\mu = xs$ satisfies

$$\frac{5}{6}LR \leq \mu \leq \frac{7}{6}LR.$$

We write μ into its three parts $(\mu^{+}, \mu^{-}, \mu^{\theta})$. By Lemma 3, we have that $x \stackrel{\text{def}}{=} (x^{+}, x^{-}, x^{\theta})$ minimizes the function

$$f_{\mu}(x^{+}, x^{-}, x^{\theta}) \stackrel{\text{def}}{=} \langle c^{+}, x^{+} \rangle + \langle c^{-}, x^{-} \rangle - \sum_{i=1}^{n} \mu_i^{+} \log x_i^{+} - \sum_{i=1}^{n} \mu_i^{-} \log x_i^{-} - \mu^{\theta} \log x^{\theta}$$

over the domain $\mathcal{P}_\eta$. The gradient of f_μ is somewhat complicated. We avoid it by considering the directional derivative at x along a feasible direction toward a lifted version of a point v such that $\mathbf{A}v = b$ and $v \geq r$. Since our domain is in $\mathcal{P}_\eta \subset \mathbb{R}^{2n+1}$, we lift v to the higher-dimensional point defined by

$$\begin{aligned}
v^- &= \min(x^-, \frac{4L\overline{R}}{\overline{t}} \cdot R), \\
v^+ &= v + v^-, \\
v^\theta &= \widetilde{b} - \sum_{i=1}^n v_i^+.
\end{aligned}$$

First, we need to get some basic bounds on $\widetilde{b}$ and c^-.

Lemma 26. *We have that $\frac{4}{5}n\overline{R} \leq \widetilde{b} \leq 3n\overline{R}$ and $c_i^- \geq 3\overline{t}/(7\overline{R})$ for all i.*

Proof. By (A.2), we have $\frac{4}{5}\overline{R} \leq \overline{x}_i^+ \leq \frac{4}{3}\overline{R}$. By the definition of $\widetilde{b}$, we have

$$\widetilde{b} = \sum_{i=1}^n \overline{x}_i^+ + \overline{R} \leq \frac{4}{3}n\overline{R} + \overline{R} \leq 3n\overline{R}.$$

Similarly, we have $\widetilde{b} = \sum_i x_{c,i}^+ + \overline{R} \geq \frac{4}{5}n\overline{R}$.

For the bound on c_i^-, recall that $c_i^- = t/\overline{x}_i^-$ with $\overline{x}^- = \overline{x}^+ - x_\circ$ and $x_\circ = \mathbf{A}^\top(\mathbf{A}\mathbf{A}^\top)^{-1}b$. Using (A.3), we have

$$x_{c,i}^- \leq \frac{4}{3}\overline{R} + R.$$

Therefore, $c^- \geq 3\overline{t}/(7\overline{R})$. $\square$

The following lemma shows that $(v^+, v^-, v^\theta) \in \mathcal{P}_\eta$.

Lemma 27. *We have that $(v^+, v^-, v^\theta) \in \mathcal{P}_\eta$. Furthermore, we have $v^\theta \geq \frac{1}{2}n\overline{R}$.*

Proof. Note that (v^+, v^-, v^θ) satisfies the linear constraints of $\mathcal{P}_\eta$ by construction. It suffices to prove that the vector is positive. Since $\overline{x}^- > 0$, we have $v^- > 0$. Since $v \geq r$, we also have $v^+ > 0$. For v^θ, we use $\widetilde{b} \geq \frac{4}{5}n\overline{R}$ (Lemma 26), $v \leq R$ and $v^- \leq \frac{4L\overline{R}}{\overline{t}} \cdot R \leq R$ to get

$$v_i^+ = v + v^- \leq 2R \tag{A.4}$$

and hence

$$v^\theta = \widetilde{b} - \sum_{i=1}^n v_i^+ \geq \frac{4}{5}n\overline{R} - \sum_{i=1}^n (v_i + v_i^-) \geq \frac{1}{2}n\overline{R}.$$

$\square$

Next, we define the path $p(\tau) = (1-\tau)(x^+, x^-, x^\theta) + \tau(v^+, v^-, v^\theta)$. Since $p(0)$ minimizes f_μ, we have that $\frac{d}{d\tau} f_\mu(p(\tau))|_{\tau=0} \geq 0$. In particular, we have

$$\begin{aligned}
0 &\leq \frac{d}{d\tau} f_\mu(p(\tau))|_{\tau=0} \\
&= \langle c, v^+ - x^+ \rangle + \langle c^-, v^- - x^- \rangle - \sum_{i=1}^n \frac{\mu_i^+}{x_i^+}(v^+ - x^+)_i - \sum_{i=1}^n \frac{\mu_i^-}{x_i^-}(v^- - x^-)_i - \frac{\mu^\theta}{x^\theta}(v^\theta - x^\theta) \\
&= \frac{\mu^\theta}{x^\theta}(x^\theta - v^\theta) + \sum_{i=1}^n (c_i - \frac{\mu_i^+}{x_i^+})(v^+ - x^+)_i + \sum_{i=1}^n (c_i^- - \frac{\mu_i^-}{x_i^-})(v^- - x^-)_i.
\end{aligned} \tag{A.5}$$

Now, we bound each term one by one. For the first term, we note that

$$\frac{\mu^\theta}{x^\theta}(x^\theta - v^\theta) \leq \mu^\theta \leq 2LR. \tag{A.6}$$

For the second term in (A.5), we have the following:

Lemma 28. *We have that* $\sum_{i=1}^n (c_i - \frac{\mu_i^+}{x_i^+})(v^+ - x^+)_i \leq 4nLR - \frac{LRr}{2\min_i x_i^+}$.

Proof. Note that

$$\begin{aligned}\sum_{i=1}^n (c_i - \frac{\mu_i^+}{x_i^+})(v^+ - x^+)_i &= \sum_{i=1}^n (c_i v_i^+ - \frac{\mu_i^+}{x_i^+} v_i^+ - c_i x_i^+ + \mu_i^+) \\ &\leq \sum_{i=1}^n c_i v_i^+ + \sum_{i=1}^n \mu_i^+ - \sum_{i=1}^n \frac{\mu_i^+}{x_i^+} v_i^+ \\ &\leq \|c\|_\infty \|v^+\|_1 + 2nLR - \frac{1}{2}\sum_{i=1}^n \frac{LRr}{x_i^+}\end{aligned}$$

where we used $\mu_i^+ \in [\frac{LR}{2}, 2LR]$ and $v_i^+ \geq v_i \geq r$ at the end. The result follows from $\|c\|_\infty \leq L$, $\|v^+\|_1 \leq 2nR$ (A.4). $\square$

For the third term in (A.5), we have the following:

Lemma 29. *We have that* $\sum_{i=1}^n (c_i^- - \frac{\mu_i^-}{x_i^-})(v^- - x^-)_i \leq \frac{1}{2}LR - \frac{\bar{t}}{8\overline{R}} \max_i x_i^-$.

Proof. Using $v^- = \min(x^-, \frac{4L\overline{R}}{\bar{t}} \cdot R)$, we have $v_i^- \leq x_i^-$. We can ignore the terms with $v_i^- = x_i^-$.

For $v_i^- < x_i^-$, we have $x_i^- \geq \frac{4L\overline{R}}{\bar{t}} R$. Using $c_i^- \geq 3\bar{t}/(7\overline{R})$ (Lemma 26), we have

$$c_i^- - \frac{\mu_i^-}{x_i^-} \geq c_i^- - \frac{\mu_i^-}{\frac{4L\overline{R}}{\bar{t}} R} \geq c_i^- - \frac{\frac{7}{6}LR}{\frac{4L\overline{R}}{\bar{t}} R} = c_i^- - \frac{7\bar{t}}{24\overline{R}} \geq \frac{\bar{t}}{8\overline{R}}.$$

Hence, we have

$$\sum_{i=1}^n (c_i^- - \frac{\mu_i^-}{x_i^-})(v^- - x^-)_i \leq \frac{\bar{t}}{8\overline{R}} \sum_{i=1}^n (v^- - x^-)_i \leq \frac{\bar{t}}{8\overline{R}} (\frac{4L\overline{R}}{\bar{t}} \cdot R - \max_i x_i^-).$$

$\square$

Combining (A.5), (A.6), Lemma 28 and Lemma 29, we have

$$\begin{aligned}0 &\leq \frac{\mu^\theta}{x^\theta}(x^\theta - v^\theta) + \sum_{i=1}^n (c_i - \frac{\mu_i^+}{x_i^+})(v^+ - x^+)_i + \sum_{i=1}^n (c_i^- - \frac{\mu_i^-}{x_i^-})(v^- - x^-)_i \\ &\leq 2LR + 4nLR - \frac{LRr}{2\min_i x_i^+} + \frac{1}{2}LR - \frac{\bar{t}}{8\overline{R}} \max_i x_i^- \\ &\leq 6nLR - \frac{LRr}{2\min_i x_i^+} - \frac{\bar{t}}{8\overline{R}} \max_i x_i^-.\end{aligned}$$

Hence, we show that (x^+, x^-, x^θ) is close to the central path at $t = LR$ implies that $\min_i x_i^+$ cannot be too small and $\max_i x_i^-$ cannot be too large

$$\frac{LRr}{2\min_i x_i^+} + \frac{\bar{t}}{8\overline{R}} \max_i x_i^- \leq 6nLR.$$

In particular, this shows the following:

Lemma 30. *We have that* $\min_i x_i^+ \geq \frac{r}{12n}$ *and* $\max_i x_i^- \leq \frac{48nL\overline{R}}{\bar{t}} \cdot R$.

Now, we are ready to prove Theorem 9.

Proof of Theorem 9. First we check $x \stackrel{\text{def}}{=} x^+ - x^- \in \mathcal{P}$. By the choice of $\overline{R}$ and $\bar{t}$, Lemma 30 shows that

$$\frac{\max_i x_i^-}{\min_i x_i^+} \leq \frac{\frac{48nL\overline{R}}{\bar{t}} \cdot R}{\frac{r}{12n}} = 576n^2 \frac{R}{r} \cdot \frac{L\overline{R}}{\bar{t}} < \eta$$

Hence, we have $x^+ - x^- > 0$ and that $\mathbf{A}(x^+ - x^-) = b$.

Next, we check $s \overset{\text{def}}{=} s^{+} - s^{\theta} \in \mathcal{D}$. Since $x \in \mathcal{P}$, we have $x_i \leq R$. Since $x_i^{-} \leq \frac{1}{2}x_i^{+}$, we have $x_i^{+} \leq 2x_i \leq 2R$. Since $x_i^{+}s_i^{+} \geq \frac{5}{6}LR$, we have $s_i^{+} \geq \frac{1}{3}L$. On the other hand, we have $x^{\theta} = \widetilde{b} - \sum_{i=1}^{n} x_i^{+} \geq \widetilde{b} - 2nR \geq \frac{1}{2}n\overline{R}$ (Lemma 26). Hence, $s^{\theta} \leq \frac{\frac{7}{6}LR}{\frac{1}{2}n\overline{R}} \leq \frac{3LR}{n\overline{R}}$. Combining both and the choice of $\overline{R}$, we have

$$\frac{s^{\theta}}{\min_i s_i^{+}} \leq \frac{\frac{3LR}{n\overline{R}}}{L/2} = \frac{6R}{n\overline{R}} < \eta$$

Hence, we have $s = s^{+} - s^{\theta} > 0$ and that $\mathbf{A}^{\top}y + s = \mathbf{A}^{\top}y + s^{+} - s^{\theta} = \mathbf{A}^{\top}y + s^{+} + \lambda = c$ (See (A.1)). $\square$

To ensure the reduction does not increase the complexity of solving linear system, we note that the linear constraint in the modified linear program is

$$\overline{\mathbf{A}} = \begin{bmatrix} \mathbf{A} & -\mathbf{A} & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

For any diagonal matrices $\mathbf{W}_1, \mathbf{W}_2$ and any scalar α, we have

$$\mathbf{H} \overset{\text{def}}{=} \overline{\mathbf{A}} \begin{bmatrix} \mathbf{W}_1 & \mathbf{0} & 0 \\ \mathbf{0} & \mathbf{W}_2 & 0 \\ 0 & 0 & \alpha \end{bmatrix} \overline{\mathbf{A}}^{\top} = \begin{bmatrix} \mathbf{A}(\mathbf{W}_1 + \mathbf{W}_2)\mathbf{A}^{\top} & \mathbf{A}w_1 \\ (\mathbf{A}w_1)^{\top} & \|w_1\|_1 + \alpha \end{bmatrix}.$$

Note that the second row and second column block has size 1. By the block inverse formula (Fact 19), $\mathbf{H}^{-1}v$ is an explicit formula involving $(\mathbf{A}(\mathbf{W}_1 + \mathbf{W}_2)\mathbf{A}^{\top})^{-1}v_{1:d}$ and $(\mathbf{A}(\mathbf{W}_1 + \mathbf{W}_2)\mathbf{A}^{\top})^{-1}\mathbf{A}w_1$. Hence, we can compute $\mathbf{H}^{-1}v$ by solving two linear systems of the form $\mathbf{A}\mathbf{W}\mathbf{A}^{\top}$ with some extra linear work.